(a) Decompose this representation in a direct sum of irreducible representations (i.e., find the multiplicities of occurrence of all irreducible representations).

(b) Do the same for the representation of G on the space of functions on the set of faces and the set of edges of the icosahedron.

Problem 4.12.6. Let $\mathbb{F}_q$ be a finite field with q elements, and let G be the group of nonconstant inhomogeneous linear transformations, $x \to ax+b$, over $\mathbb{F}_q$ (i.e., $a \in \mathbb{F}_q^\times, b \in \mathbb{F}_q$). Find all irreducible complex representations of G, and compute their characters. Compute the tensor products of irreducible representations.

Hint: Let V be the representation of G on the space of functions on $\mathbb{F}_q$ with sum of all values equal to zero. Show that V is an irreducible representation of G.

Problem 4.12.7. Let $G = SU(2)$ (the group of unitary 2×2 matrices with determinant 1), and let $V = \mathbb{C}^2$ be the standard 2-dimensional representation of $SU(2)$. We regard V as a real representation, so it is 4-dimensional.

(a) Show that V is irreducible (as a real representation).

(b) Let $\mathbb{H}$ be the subspace of $\mathrm{End}_{\mathbb{R}}(V)$ consisting of endomorphisms of V as a real representation. Show that $\mathbb{H}$ is 4-dimensional and closed under multiplication. Show that every nonzero element in $\mathbb{H}$ is invertible, i.e., $\mathbb{H}$ is an algebra with division.

(c) Find a basis $1, i, j, k$ of $\mathbb{H}$ such that 1 is the unit and

$$i^2 = j^2 = k^2 = -1,\ ij = -ji = k,\ jk = -kj = i,\ ki = -ik = j.$$

Thus we have that Q_8 is a subgroup of the group $\mathbb{H}^\times$ of invertible elements of $\mathbb{H}$ under multiplication.

The algebra $\mathbb{H}$ is called the **quaternion algebra**, and its elements are called **quaternions**. Quaternions were discovered by W. R. Hamilton in 1843 (see Section 4.13).

(d) For $q = a + bi + cj + dk$, $a, b, c, d \in \mathbb{R}$, let $\bar{q} = a - bi - cj - dk$ and $||q||^2 = q\bar{q} = a^2 + b^2 + c^2 + d^2$. Show that $\overline{q_1 q_2} = \bar{q}_2 \bar{q}_1$ and $||q_1 q_2|| = ||q_1|| \cdot ||q_2||$.